TECHNICAL PERFORMANCE REPORT / VERSION 1.0

Measured performance. Traceable evidence.

An evaluation of VedAI's Sanskrit retrieval, source preservation, structured answer checks and application performance.

85.4%

Retrieval Recall@10 | Vocabulary-expanded baseline

Average coverage of expected source segments within the first ten retrieved chunks, measured across 48 natural-language questions in four language forms.

0.631

Mean reciprocal rank

Natural-question retrieval

15 / 15

Exact-quote targets found first

One target for each indexed work

20 / 20

Answer check cases passed

Latest recorded mechanical checks

EVALUATION SCOPE

This report presents recorded development measurements. Retrieval percentages describe source recovery; answer checks describe structural conformance and source integrity. Each result is tied to its dataset and evaluation procedure.

A source-grounded reading system

VedAI combines natural-language query planning with retrieval from the SARIT Ayurveda collection. Its reading interface presents original Sanskrit, Devanagari rendering, translations and stored book, chapter and verse locators alongside the generated answer.

INDEXED COLLECTION

15 Ayurveda works
26,585 retrieval chunks
111,195 complete source passages

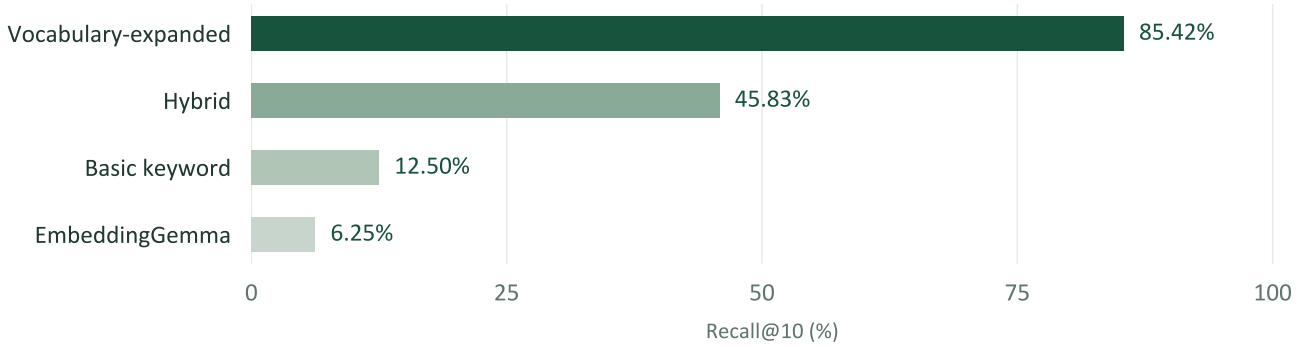
REPORT CONTENTS

Retrieval performance and corpus provenance.
Answer checks and operational measurements.
Evaluation methodology and source records.

01 / RETRIEVAL EVALUATION

Finding the relevant source.

Vocabulary-expanded retrieval achieved the highest recall and ranking score among the four evaluated configurations.



Retrieval configuration	Recall@10	MRR
Vocabulary-expanded	**85.42%**	**0.631**
Hybrid	45.83%	0.228
Basic keyword	12.50%	0.095
EmbeddingGemma	6.25%	0.026

Performance by query language

Language form	Questions	Recall@10	MRR
English	12	75.00%	0.516
Hindi	12	83.33%	0.541
Romanized Sanskrit	12	91.67%	0.701
Devanagari Sanskrit	12	91.67%	0.764

WHAT THE SCORES MEASURE

Recall@10 is the average fraction of expected source segments recovered within ten chunks. MRR averages the reciprocal rank of the first relevant chunk, assigning zero when none is found. Higher scores indicate better retrieval.

The 48 questions represent 12 topics in four language forms across three core works. Book filters were supplied. These baseline measurements precede AI query enhancement and the later avagraha normalization update.

Records: [R1] Retrieval evaluation files and [R2] Dataset definition.

02 / COLLECTION AND PROVENANCE

Preserving the original evidence.

The complete pinned SARIT download contains 110 files, including 88 XML files. The processed Ayurveda collection comprises the following 15 works.

145,600
Extracted source segments

26,585
Searchable retrieval chunks

111,195
Complete source passages

Indexed work	Segments	Chunks
Charaka Samhita	10,866	775
Sushruta Samhita	10,486	686
Ashtanga Hridaya	11,097	529
Ashtanga Sangraha	22,971	786
Bhela Samhita	2,821	215
Chakradatta	4,788	295
Chakrapani's Ayurvedadipika	27,010	5,822
Arunadatta's Sarvangasundara (with base text and Ayurvedarasayana)	27,836	13,260
Rasa Ratna Samuccaya with commentaries	6,639	863
Ayurvedasutram	466	16
Siddhayoga	3,861	243
Ugraditya's Kalyanakaraka	7,866	1,994
Rasahrdayatantra	2,865	750
Rasaratnakara, Rasayanakhanda	1,302	191
Rasarnava	4,726	160

EXTRACTION AND INDEX CHECKS

110 source files verified
119,774 citation spans verified
Chunk coverage and index identities verified

PASSAGE PRESERVATION

119,407 constituent segments checked
8,167 joined source units
Stored text and locators matched throughout

Exact-quote retrieval recovered all 15 target passages at rank 1. These checks verify text preservation, traceability and source-location resolution.

Records: [R3] Corpus inventory, [R4] Extraction validation and [R5] Passage integrity. Pinned SARIT revision: 1eac9ee0b055.

03 / ANSWER PIPELINE VERIFICATION

Consistent answers. Explicit source boundaries.

All 20 development cases passed their latest recorded mechanical checks across implementation iterations.

10
Clinical case examples

10
Reference question examples

20 / 20
Complete check sets passed

Check	Acceptance criterion	Passed
Evidence available	A retrieved evidence packet is present.	**20 / 20**
Source preservation	Sanskrit, verse labels, sections and references match storage.	**20 / 20**
Generation completion	A completed structured answer is returned.	**20 / 20**
Question structure	The answer uses an accepted question type.	**20 / 20**
Clinical priority order	The priority statement appears first in designated urgent cases.	**4 / 4**
Case findings	Patient-case answers include structured findings.	**11 / 11**
Citation separation	General and clinical context carry no borrowed corpus citations.	**20 / 20**
Translation presence	Each cited source has a generated translation.	**20 / 20**

Additional source and translation checks

SOURCE NAVIGATION

All 19 recorded guide ranges resolve their expected verse numbers against the pinned SARIT edition.

TRANSLATION INTEGRATION

Two recorded Sol corpus-answer checks produced five translated source passages in total, with source text and locators preserved.

> **INTERPRETATION OF THE PASS RESULTS**
>
> The pass counts describe the stated acceptance criteria. Translation checks establish presence and source binding. Conditional checks use only applicable cases. The 20 examples are development regression cases drawn from the supplied Hindi materials.

The recorded set combines each case's latest result. The Sama/Nirama example accepts disease or procedure structure because its question includes both classification and treatment eligibility.

Records: [R6] Clinical check summary, [R7] Source-guide checks and [R8] Translation integration checks.

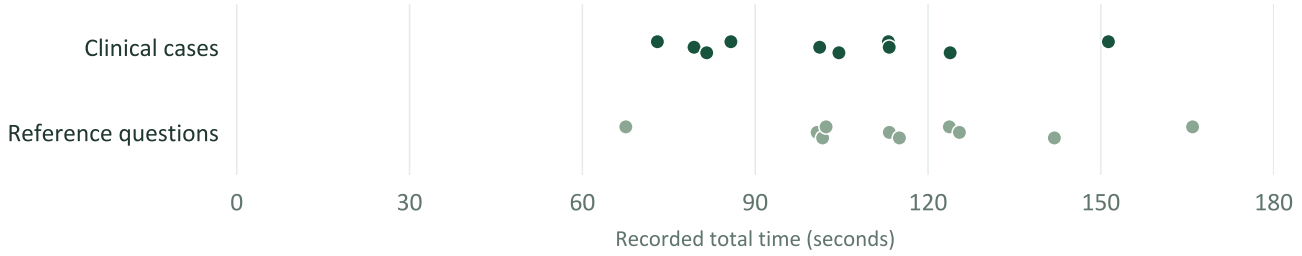
04 / TIMING AND APPLICATION CHECKS

Execution performance.

Recorded timing separates local retrieval from the complete planning and answer-generation workflow.

108.83 s

Median total time across 20 recorded generated-answer development cases



Recorded workload	Runs	Median	Mean	Observed range
Local retrieval only	20	0.39 s	0.43 s	0.18 to 1.14 s
Planning and generated answer	20	108.83 s	109.23 s	67.54 to 165.92 s

Timing includes the work performed by each respective check harness. Generated-answer observations are drawn from the latest recorded case runs across iterations.

HOSTED RETRIEVAL OBSERVATION

A GCP retrieval replay returned 40 exact stored passages in 2.738 seconds after the source-locator index update. This is a single observed run.

Application verification

57 / 57

Local automated tests passed
Verification recorded 24 September 2026

8 / 8

Reading-interface viewport sizes passed
Desktop, tablet, mobile and landscape

Browser checks covered contained scrolling, source navigation, Devanagari display, copying and preferences. Session checks passed for creation, renaming, deletion, history persistence, context isolation, job recovery, updates across tabs and unavailable browser storage.

Records: [R6] Generated-answer timings, [R9] Retrieval and hosted observations, and [R10] Report verification record.

05 / METHODOLOGY AND RECORD REGISTER

Defined measures. Reproducible records.

Retrieval sample	The draft dataset contains 68 cases: 48 natural questions, 15 exact-quote targets and five unanswerable prompts. Natural questions cover 12 topics across four language forms and three works.
Evaluation design	Relevance labels were authored during development. Work filters were supplied, and vocabulary was refined after inspecting initial results. Development and validation topic groups are separated by work. The scores describe this development dataset.
Recall and ranking	Recall@10 averages expected-segment coverage over answerable questions. MRR averages 1 divided by the first relevant rank, or zero when no relevant chunk appears. Unanswerable prompts are excluded from both measures.
Answer checks	The 20-case summary records the latest mechanical result for each example across iterations. Criteria include completion, structure, source preservation, translation presence and applicable priority or finding requirements.
System versions	Retrieval baselines precede AI enhancement and later text normalization. Recorded Sol answer checks use medium reasoning. Application verification in this report refers to revision ea1e2f2.
Timing and rounding	Means and medians are calculated from recorded per-case durations. Workloads and environments are identified alongside each result. Displayed percentages and timings are rounded; source files retain their recorded values.

Source register

[R1] Retrieval measurements
eval-expanded.json; eval-hybrid.json; eval-lexical.json; eval-semantic.json

[R2] Dataset and evaluation procedure
eval/README.md; eval/questions.jsonl; vedai/evaluation.py

[R3] Collection inventory
reports/corpus-summary.json

[R4] Extraction and index checks
reports/validation.json

[R5] Passage preservation
reports/passages-integrity.json

[R6] Structured answer results
reports/clinical-summary.json; tests/live_clinical.py

[R7] Source-guide resolution
reports/source-guide-checks.json

[R8] Translation integration
reports/translation-checks.json

[R9] Execution-time observations
clinical-retrieval-checks.json; clinical-deployment.json

[R10] Current application verification
reports/performance/verification.json

Document control
VedAI Performance & Evaluation Report, version 1.0. Prepared 24 September 2026.
Corpus: SARIT revision 1eac9ee0b055c8d11147edac4a75c76008ccc363.
Report generated from repository records. Application checks use retrieval-only requests and browser fixtures.